



GE MRI

MRI IMAGE RECONSTRUCTION

From Raw Data to Final Image

MRI does not capture images directly. It measures signals in k-space and uses mathematics to reconstruct the images we see.



THE BIG IDEA

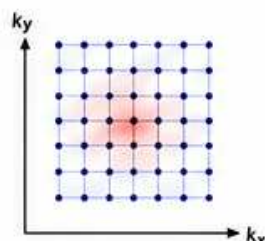
MRI reconstructs an image through a series of well-defined steps: from signal sampling in k-space to a final image optimized for clinical use.

THE BASICS

- 1 MRI measures signals from hydrogen protons after RF excitation.
- 2 Signals are encoded with gradients and stored as data points in k-space.
- 3 The raw k-space data is converted to an image using the Fourier Transform.
- 4 The image is processed and optimized for diagnostic use.
- 5 Many factors influence image quality at each step.

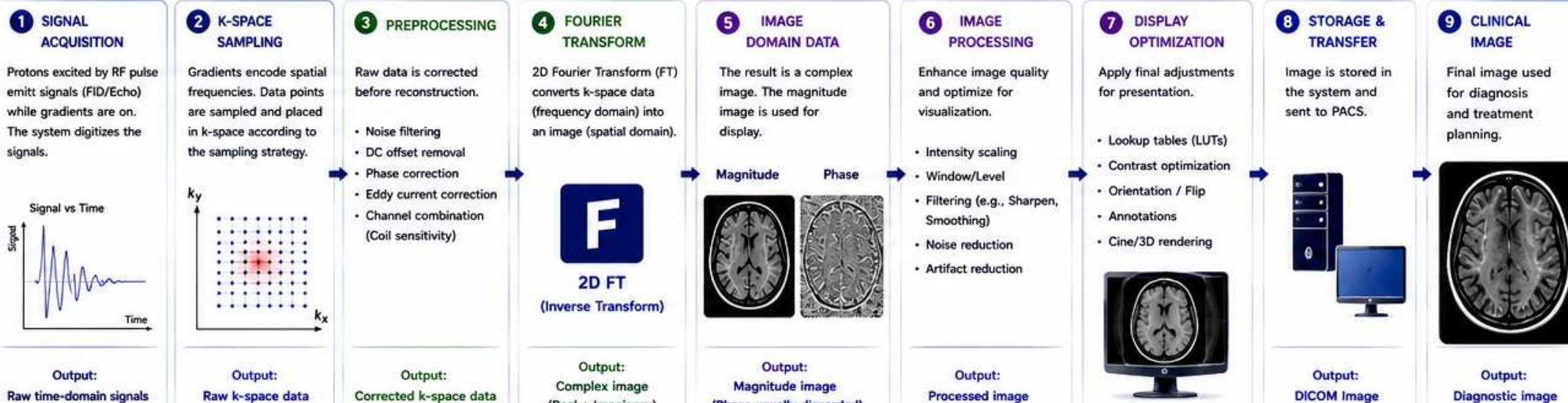
WHAT IS K-SPACE?

K-space (frequency space) is a matrix that contains all the frequency information needed to build an image.

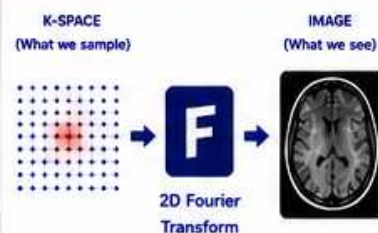


- Center = Contrast
- Outer = Detail

THE MRI IMAGE RECONSTRUCTION PIPELINE



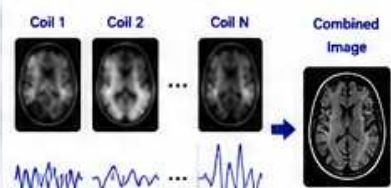
K-SPACE TO IMAGE (THE BIG PICTURE)



K-space contains all spatial frequency information. Fourier Transform converts it into a spatial image.

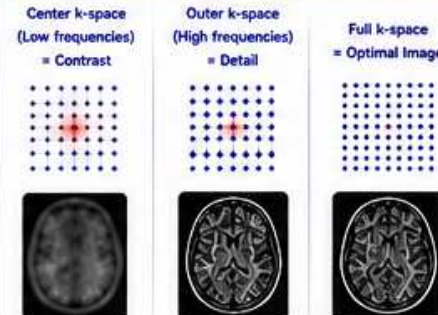
COIL SENSITIVITY & COMBINATION

Multiple receiver coils capture the same anatomy with different sensitivity profiles.



Coil sensitivities are used to combine signals and improve SNR and reduce artifacts (e.g., SENSE).

K-SPACE & IMAGE RELATIONSHIP



More center data = better contrast.
More outer data = better detail and sharpness.

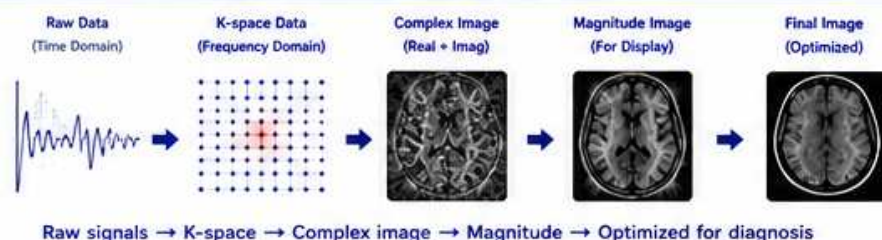
COMMON RECONSTRUCTION METHODS

Method	Description	Key Use
DFT (FFT)	Standard Fourier Transform	Most routine imaging
Filtered Back Projection	Used in CT, not typical in MRI	CT imaging
Parallel Imaging (SENSE)	Uses coil data to speed up scans	Reduces scan time
Partial Fourier	Samples a portion of k-space	Faster scans (phase encode)
Compressed Sensing	Reconstructs from undersampled data	High acceleration with sparsity

WHAT AFFECTS RECONSTRUCTION QUALITY?

- SNR**
Higher SNR = better image quality
- K-space Coverage**
More complete data = better resolution
- Sampling Strategy**
Affects artifacts, scan time, and resolution
- Motion**
Causes phase errors and ghosting
- System & Hardware**
Gradients, coils, calibration, stability
- Reconstruction Parameters**
Filters, matrix size, bandwidth, algorithms

TYPES OF DATA THROUGH RECONSTRUCTION



RECONSTRUCTION TRADE-OFFS

Better Image Quality
More k-space data
Higher resolution
Higher SNR
Less artifact
Longer scan time



Faster Imaging
Less k-space data
Lower resolution
Lower SNR
More artifact
Shorter scan time

The goal is to find the right balance for the clinical task.

KEY TAKEAWAYS

- ✓ MRI measures signals, not images.
- ✓ K-space sampling captures spatial frequency information.
- ✓ Fourier Transform converts k-space data into an image.
- ✓ Image processing and optimization enhance diagnostic value.
- ✓ Understanding each step helps troubleshoot and optimize images.



SUMMARY

Complete pipeline from raw data to final diagnostic image.



GE MRI CLINICAL ENGINEER HANDBOOK

Better Understanding. Better Troubleshooting. Better Patient Care.

K-SPACE & IMAGE RECONSTRUCTION